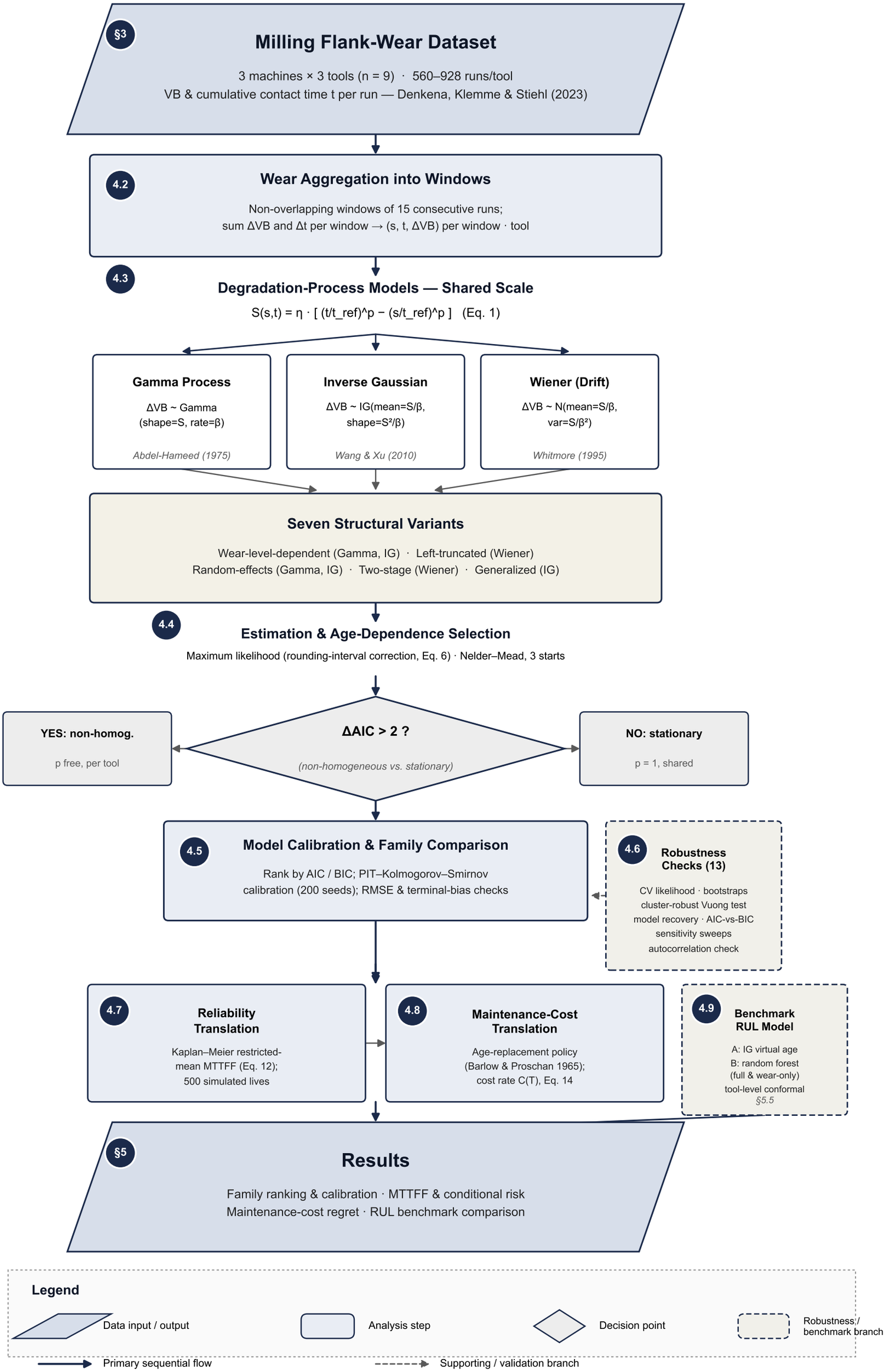

§3
Milling Flank-Wear Dataset
3 machines × 3 tools (n = 9) · 560–928 runs/tool
VB & cumulative contact time t per run — Denkena, Klemme & Stiehl (2023)
4.2
Wear Aggregation into Windows
Non-overlapping windows of 15 consecutive runs;
sum ΔVB and Δt per window → (s, t, ΔVB) per window · tool
4.3
Degradation-Process Models — Shared Scale
S(s,t) = η · [(t/t_ref)^p − (s/t_ref)^p] (Eq. 1)
Gamma Process
ΔVB ~ Gamma
(shape=S, rate=β)
Abdel-Hameed (1975)
Inverse Gaussian
ΔVB ~ IG(mean=S/β,
shape=S²/β)
Wang & Xu (2010)
Wiener (Drift)
ΔVB ~ N(mean=S/β,
var=S/β²)
Whitmore (1995)
Seven Structural Variants
Wear-level-dependent (Gamma, IG) · Left-truncated (Wiener)
Random-effects (Gamma, IG) · Two-stage (Wiener) · Generalized (IG)
4.4
Estimation & Age-Dependence Selection
Maximum likelihood (rounding-interval correction, Eq. 6) · Nelder–Mead, 3 starts
YES: non-homog.
p free, per tool
ΔAIC > 2 ?
(non-homogeneous vs. stationary)
NO: stationary
p = 1, shared
4.5
Model Calibration & Family Comparison
Rank by AIC / BIC; PIT–Kolmogorov–Smirnov
calibration (200 seeds); RMSE & terminal-bias checks
4.6
Robustness Checks (13)
CV likelihood · bootstraps
cluster-robust Vuong test
model recovery · AIC-vs-BIC
sensitivity sweeps
autocorrelation check
4.7
Reliability Translation
Kaplan–Meier restricted-
mean MTTFF (Eq. 12);
500 simulated lives
4.8
Maintenance-Cost Translation
Age-replacement policy
(Barlow & Proschan 1965);
cost rate C(T), Eq. 14
4.9
Benchmark RUL Model
A: IG virtual age
B: random forest
(full & wear-only)
tool-level conformal
§5.5
§5
Results
Family ranking & calibration · MTTFF & conditional risk
Maintenance-cost regret · RUL benchmark comparison
Legend
Data input / output
Analysis step
Decision point
Robustness / benchmark branch
Primary sequential flow
Supporting / validation branch